PRAMAAN

A 10-year architecture that centralises trust, federates delivery and makes a stolen paper worthless

Hybrid by function: one statutory authority holds standards, keys and audit, exam bodies and states deliver, and item banks shrink what any insider can steal.

India loses about ₹3,317 Cr a year to cancelled exams. PRAMAAN cuts that by 80% for ₹135 a candidate sitting and returns ₹3.14 per rupee over ten years.

TEAM IDEAFORGE

Team Members	Institution	Role
Harsh Kawatra	Delhi Technological University	Team Leader
Gursimran Kaur	Guru Gobind Singh Indraprastha University	Member
Dayita Arora	Ramjas College, University of Delhi	Member

THE CASE IN FIVE NUMBERS

22.7 lakh
NEET-UG 2026 candidates whose exam was cancelled after setters and translators leaked it

41 leaks
in 15 states over five years hit 1.4 crore job seekers (Indian Express, 2024)

87%
of candidates in attributed cancellations lost the paper before it reached a centre

39
permanent posts at the NTA, beside 197 contract and outsourced staff (Jul 2026)

96x
smaller blast radius under PRAMAAN: 23,646 candidates per breach, not 22.7 lakh

Maudrik logo

Team IdeaForge logo

Source: Maudrik v7.0 problem statement | Wikipedia and CBI on NEET-UG 2026 | Indian Express (Feb 2024) | Rajya Sabha reply (22 Jul 2026) | Team IdeaForge PRAMAAN model and 49-incident dataset

A federated architecture that centralises trust and shrinks the prize cuts India's ₹3,317 Cr yearly breach loss by 80% for ₹135 a candidate sitting

BREACHES COST ₹3,317 CR A YEAR

₹3,317 Cr

lost a year to cancelled exams (base case)

38.6 lakh candidates a year sit an exam that is later cancelled

87% of them lost the paper before it reached a centre

31 of 49 coded breaches were insider jobs

WHY THE OBVIOUS FIX FALLS SHORT

2026 response	What it misses
NEET-UG on CBT from 2027	2026 leak came from setters, before delivery
Fines up to ₹10 Cr	Needs detection: 52 FIRs under the 2024 Act
One CBT platform	One vendor becomes one point of failure
AI cameras in exam halls	Watches candidates, not the vault

OUR ANSWER: PRAMAAN

1. Shrink the prize: item banks and 96 parallel forms, so no full paper exists until an hour before
2. Split the keys: 3-of-5 release across independent bodies; no one person can open a paper
3. Centralise trust, federate delivery: one authority audits, 36 states and exam bodies deliver

ANSWERS TO THE SIX QUESTIONS

Question	Our answer
1 Central vs federal	Hybrid by function: central trust, federated delivery
2 Cut the four risks	Item banks, split keys, sealed chain, zero-trust nodes
3 AI and biometrics	Encryption is the backbone; AI flags, humans decide
4 Liability	6-hour clock, uncapped vendor liability, named officer
5 Justified spend	₹673.5 Cr a year: 20% of the loss, BCR 3.14
6 Access and cost	₹135 a sitting; reserved-group waivers from Union share

WHAT ₹7,092 CR BUYS IN TEN YEARS

80% less expected loss from 2031

3.14 rupees back per rupee

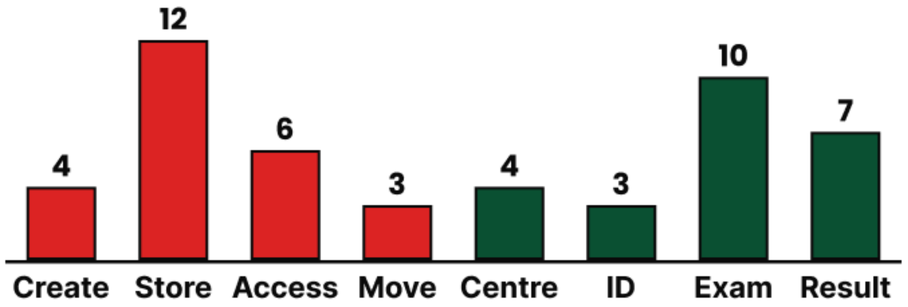
96x smaller blast radius

Bear case still returns 2.28; counting cash alone, 1.31.

Source: Indian Express (Feb 2024) | NTA, PIB and Wikipedia on NEET-UG 2026 | Rajya Sabha (Jul 2026) | PRS (Jul 2026) | NTA CBT tender (Mar 2026) | Team IdeaForge PRAMAAN model and 49-incident dataset

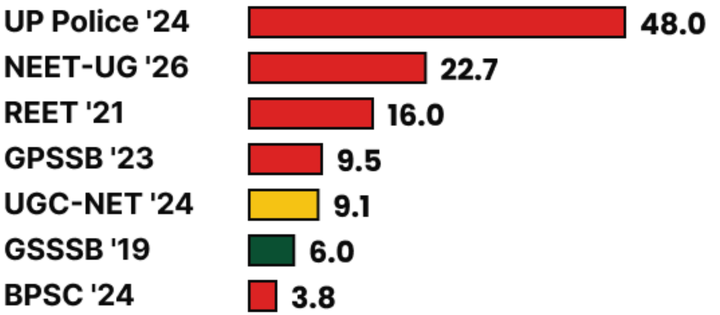
India's leaks start before the exam hall: 87% of candidates in attributed cancellations lost their paper at creation, storage or transport

25 OF 49 BREACHES START PRE-CENTRE



Red: before a centre (25 of 49 incidents)

87% LOST BEFORE THE CENTRE (LAKH)



Pre-centre: 100.7 of 115.8 lakh (87%)

ONE PAPER, MANY HANDS

A NEET-UG paper is one object shared by 22.7 lakh candidates. Before the exam it passes setters, translators, a press or server, a vault, transporters, a strong room and a centre.

Stronger locks raise the price of a leak. They do not shrink the prize.

EIGHT HANDS TOUCH THE PAPER; FIVE BREACHED IT BEFORE THE HALL

Setter NEET-UG '26	**Translator** NEET-UG '26	**Press** BPSC '24	**Vault** REET '21
Transport UP Police '24	**Strong room** NEET-UG '24	**Centre** GSSSB '19	**Evaluator** TRB '17

Red hands sit before the centre: 25 of 49 incidents and 87% of cancelled candidates

WHY MORE RULES HAVE NOT WORKED

1. Radhakrishnan's 101 fixes (Oct 2024): print-at-centre still unused by May 2026
2. NTA runs crores of tests on 39 permanent posts; 197 staff are contract or outsourced
3. The CBT tender lets vendors subcontract biometrics and fines a CCTV blind spot ₹200
4. UPSC: no confirmed leak reported since 1991; its setters do not know which items are used

Source: Team IdeaForge 49-incident dataset, 2015 to 2026 (Wikipedia list; CBI, STF, SIT and EOU findings via Careers360, ThePrint, Hans India) | Supreme Court (Jul 2024) | Rajya Sabha (Jul 2026) | Vajiram and Ravi (2026)

Insiders drive 31 of 49 coded breaches, and the two highest risks, scored 20 of 25, sit in question creation and storage rather than the exam hall

SEVEN CRITICAL CELLS, SIX BEFORE THE HALL

Stage	Insider	Physical	Cyber	Operations
Question creation	20	5	10	6
Question storage	20	15	15	8
Personnel access	16	6	12	9
Transmission	15	12	8	6
Exam centre	12	9	6	6
Authentication	4	6	4	4
Examination	9	8	16	12
Evaluation & results	9	2	8	9

≥15 critical | 10 to 14 high | 6 to 9 medium | ≤5 low

Priority = Likelihood (1 to 5, from incident frequency) x Impact (1 to 5, from candidates exposed). 15 or more is critical.

By risk type: Insider 31, Cyber 8, Physical 7, Operations 3

TOP 10 RISKS WITH 1:1 MITIGATION

#	Risk (evidence)	Mitigation in PRAMAAN	Score
1	Press or vault insider copies it (BPSC '24)	No full paper before T−60 min; per-copy watermark	20
2	Setter or translator leaks items (NEET '26)	20x item bank, 5% cap each, coaching conflict bar	20
3	Remote access to CBT nodes (JEE Main '21)	Locked node image, no remote ports, network SOC	16
4	Official sells access (TSPSC '23)	Just-in-time access, split keys, vetting, rotation	16
5	Stored papers stolen (HP Police '22)	Smart-lock boxes, GPS, tamper-evident seals	15
6	Server or file theft (MPPEB '21)	Air-gapped vault, HSM encryption, data-loss control	15
7	Logistics staff open sealed boxes (UP '24)	Encrypted e-delivery, print at centre, smart locks	15
8	Platform crash or false alarm (SSC '25)	Vendor cap, buffer nodes, 24-hour leak check	12
9	Insider misuses admin access (TSPSC '23)	Hardware keys, zero-trust sessions, access logs	12
10	Centre staff open strong room (NEET '24)	Keys at centre T−30 min, dual custody, CCTV alert	12

Source: Team IdeaForge risk assessment (likelihood from the 49-incident dataset; impact from candidates exposed) | NTA CBT tender (Mar 2026) | CBI (JEE Main 2021) | UP STF (2024) | Bihar EOU (2024) | Hans India (TSPSC 2023)

Hybrid by function: one statutory authority holds standards, keys and audit, while delivery stays federated across exam bodies, 36 states and capped vendors

Q1: ONLY HYBRID PASSES ALL FIVE TESTS

Criterion	Central	Federal	Hybrid
Single failure point	High	Low	Low
Common standards	Strong	Weak	Strong
Uses state capacity	Weak	Strong	Strong
Blast radius	22.7 lakh	Varies	23,646
Response speed	Slow	Uneven	6 hours

THE PRAMAAN ARCHITECTURE: THREE PLANES

TRUST PLANE central	Item bank vault	3-of-5 split keys	National standard	CERT-Exam SOC
ASSURANCE PLANE independent	NEIAA accredits	Audits, red teams	Incident command	Public reports
DELIVERY PLANE federated	NTA, UPSC, SSC, RRB	36 state cells	Vendors, 40% cap	2,500 secure labs

ONE ACCOUNTABLE OWNER PER STAGE (RACI)

Stage	NEIAA	Body	CERT	State cell	Vendor	Police	Auditor
Creation	C	A/R	·	·	·	I	C
Storage	R	A	C	·	·	I	C
Personnel access	A	R	C	R	R	C	C
Transmission	A	R	R	C	R	I	I
Exam centre	C	A	C	R	R	C	I
Authentication	C	A	I	C	R	I	I
Examination	I	A	R	R	R	C	I
Evaluation	C	A/R	C	·	C	I	R

RISK TIERS AND BINDING RULES

Tier	Which exams	Controls
Tier 1	5 lakh+ or national rank	Full stack: bank, keys, SOC
Tier 2	50,000 to 5 lakh	Bank, watermark, vendor audit
Tier 3	Under 50,000	Custody log and watermark

- Vendor cap: no firm above 40% of CBT volume
- Coaching bar: setters 3 years clear of coaching

Source: Team IdeaForge governance design | PEA 2024 and Amendment 2026 (PRS) | Radhakrishnan Committee (Oct 2024) | NTA CBT tender (Mar 2026) | CERT-In Directions (Apr 2022) | NIST SP 800-207 | KNEC Kenya (2016)

Item banks, split keys and per-copy watermarks shrink what one NEET-UG compromise exposes from 22.7 lakh candidates to 23,646

THE MONEYSHOT: WHAT ONE COMPROMISE CAN REACH

22,70,000 → 23,646
NEET-UG candidates exposed: today vs PRAMAAN

96x
smaller blast radius per breach

67% → 2.5%
of a paper one insider can expose

-₹5.4 Cr
expected value of a leak to its seller

EVERY STAGE GETS PREVENT, DETECT AND RESPOND

Stage	Prevent	Detect	Respond
Creation	20x bank, 5% cap each	Item-overlap scans	Retire exposed items
Storage	Air-gapped HSM vault	Access-log alerts	Re-key, rebuild forms
Personnel	Vetting, conflict bar	Behaviour analytics	Suspend, investigate
Transmission	Encrypted, keys at T−60	Tamper seals, GPS	Switch to reserve form
Exam centre	Hardened lab, jammers	CCTV analytics flags	Cancel one centre only
Authentication	1:1 face match	Liveness check	Alternate-ID desk
Examination	Locked nodes, 96 forms	Network SOC, logs	Re-test the one form
Evaluation	Encrypted responses	Answer-similarity AI	Human panel decides

ENCRYPTION

Role: backbone. HSM vault, 3-of-5 keys, encrypted answers.

Limit: no body holds a whole key.

BIOMETRICS

Role: the gate. 1:1 face match with a liveness check.

Limit: deleted 90 days after results.

AI ANALYTICS

Role: analyst. Flags answer similarity and anomalies.

Limit: a human panel decides.

SURVEILLANCE

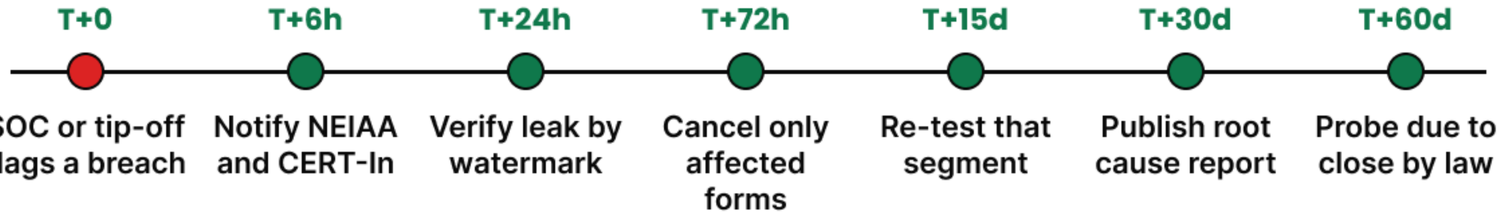
Role: proportionate. Wired CCTV in rooms and vaults.

Limit: no internet shutdowns.

Source: Team IdeaForge PRAMAAN model (96 forms = 6 days x 2 shifts x 8 forms; bank 20x paper) | Wikipedia and CBI on NEET-UG 2026 (120 of 180) | College Board | UPSC practice | NIST SP 800-207 | DPDP Rules 2025

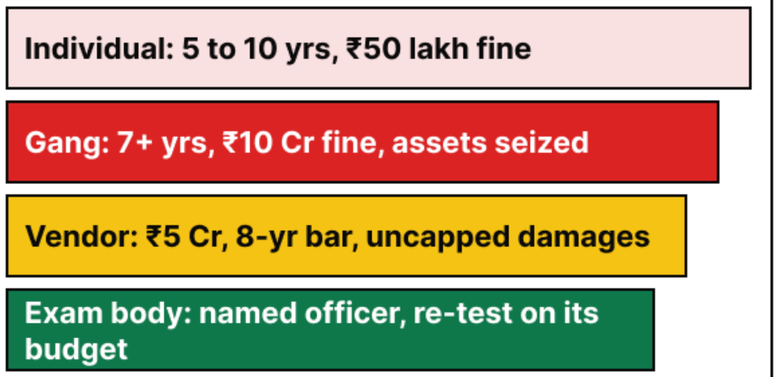
Liability that follows the breach: a six-hour clock, uncapped vendor damages and a leak whose expected value to its seller falls from +₹14 Cr to −₹5.4 Cr

FROM DETECTION TO PUBLIC REPORT IN 30 DAYS



In 2024, UGC-NET was cancelled for 9.08 lakh on an unverified screenshot; in 2026, NEET-UG took nine days to cancel.

WHO PAYS: THE LIABILITY LADDER



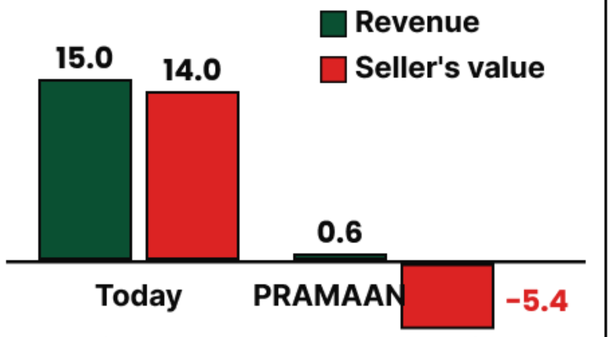
FIVE CONTRACT GAPS, FIVE FIXES

Lever	Today	PRAMAAN
Vendor duty	NTA disowns (cl. 11.3)	Strict for subcontracts
Bank guarantee	10% of exam value	15% plus insurance
Blind spot	₹200 penalty	Centre de-accredited
Breach notice	No fixed clock	6 hours to NEIAA
Public report	Rarely published	Root cause in 30 days

AUDIT CALENDAR

Audit	How often
Readiness check	Each cycle
NESS compliance	Annual
Red team	Twice a year
Answer forensics	Every exam
CAG audit	Every 3 years

A LEAK STOPS PAYING (₹ CR)

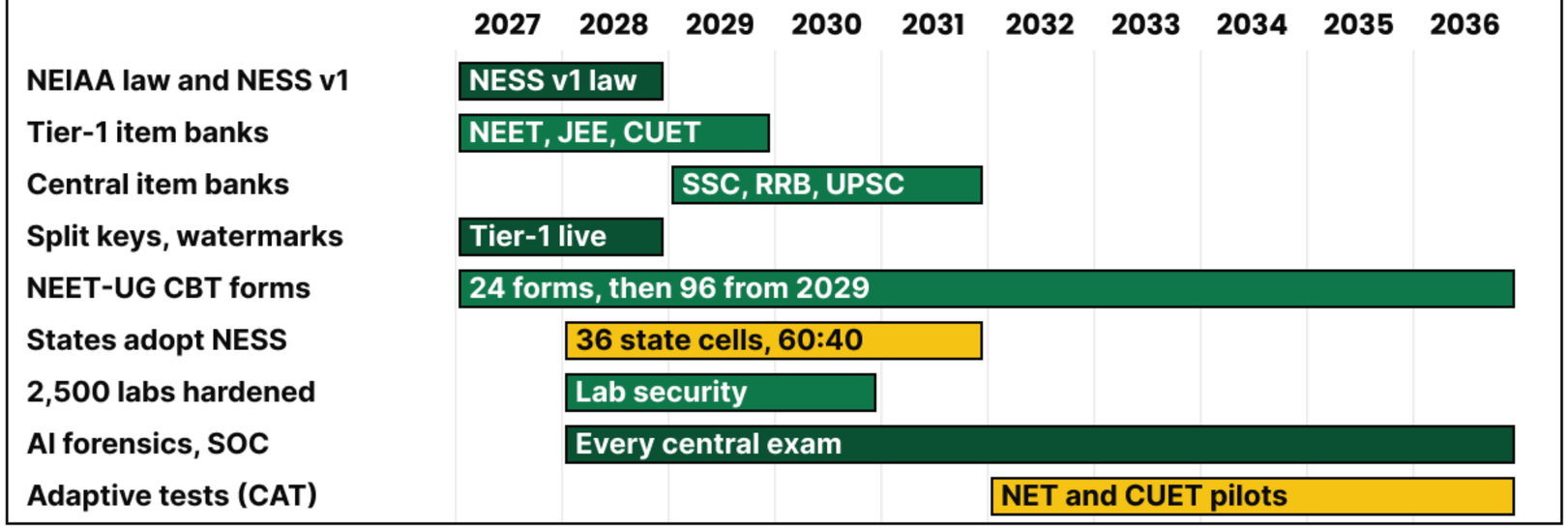


Detection odds: 10% now, 60% after

Source: PEA 2024 and Amendment 2026 (PRS, Jul 2026) | CERT-In Directions (Apr 2022) | NTA CBT tender cl. 9 to 12 (Mar 2026) | Supreme Court (Jul 2024) | CBI closure, UGC-NET (ThePrint) | PRAMAAN model (A16)

Ten years in three horizons: every Tier-1 exam traceable by 2028, national coverage by 2031, and the 80% loss cut held through 2036

NINE WORKSTREAMS, ALL LIVE BY 2032



THREE HORIZONS AND GATES

2027 TO 2028: STABILISE

Gate: zero Tier-1 cancellations; every Tier-1 copy traceable

2029 TO 2031: SCALE

Gate: blast radius under 25,000; 80% of state volume accredited

2032 TO 2036: EMBED

Gate: 80% loss cut held; cost at or under ₹135 a sitting

80% LOSS CUT FROM 2031, HELD TO 2036

KPI	2026	2028	2031	2036
Loss cut vs 2026	0%	28%	80%	80%
Blast radius, NEET-UG	22.7 lakh	94,583	23,646	23,646
Tier-1 exams traceable	0%	100%	100%	100%

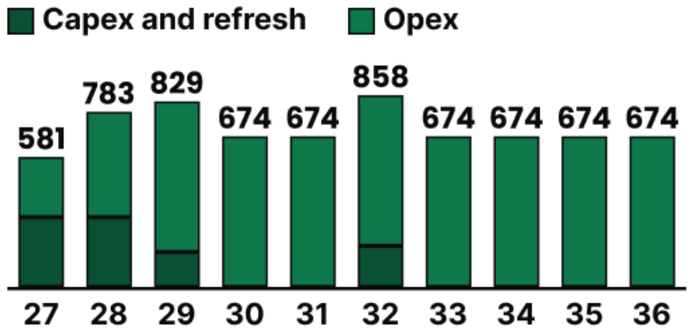
ACCESS AND AFFORDABILITY (Q6)

- ₹50 levy; SC, ST and PwD waivers paid from the Union share
- Centre within 50 km; mobile centres for remote areas
- Aadhaar optional: passport, DigiLocker and voter ID accepted
- No internet shutdowns; jammers only inside exam rooms

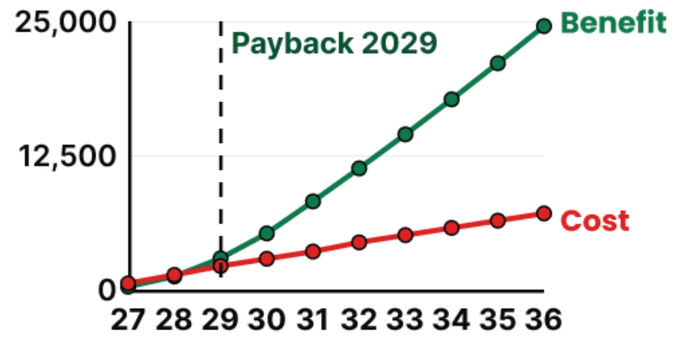
Source: Team IdeaForge roadmap and model (coverage path A10) | Radhakrishnan Committee (Oct 2024) | MoE and NTA on NEET-UG CBT from 2027 (Jul 2026) | Nilekani task force (Jul 2026) | PEA Amendment 2026 | DPDP Rules 2025

₹7,092 Cr over ten years returns ₹3.14 per rupee, clears 1.0 even on cash alone, and scores 21 of 24 on resilience against 7 for today's model

COST PEAKS AT ₹858 CR (₹ CR)



BENEFITS OVERTAKE COST IN 2029



EVERY CASE CLEARS 1.0, EVEN CASH ONLY

Case (₹ Cr)	NPV	BCR	Payback
Bear	6,092	2.28	2030
Base	10,211	3.14	2029
Bull	12,317	3.58	2028
Cash only	1,496	1.31	2032

Spend is 55% of the Gordon-Loeb ceiling of ₹1,220 Cr

PRAMAAN 21 OF 24, TODAY'S MODEL 7 (0 TO 3 PER TEST)

Test	Fragmented today	Central CBT only	PRAMAAN
Insider at creation	None	Weak	Strong
Storage and transport	None	Partial	Strong
Centre and remote access	Weak	Weak	Partial
Blast radius per breach	None	Weak	Strong
Vendor concentration	Weak	None	Strong
Detection and liability	None	Weak	Strong
Privacy proportionality	Partial	Weak	Partial
Cost and access	Strong	Weak	Partial
Total	7 / 24	8 / 24	21 / 24

TRADE-OFFS WE ACCEPT

1. Cost: ₹135 a sitting, 7.9% of the NEET-UG fee
2. Psychometrics: parallel forms need published equating
3. Privacy: face match only, deleted in 90 days, DPIA
4. A new central vault, guarded by 3-of-5 split keys
5. Federal consent: states join through 60:40 funding

Source: Team IdeaForge PRAMAAN model (T2 to T6; PV at 8% real; 2026 prices) | Indian Express (Feb 2024) | NTA finances (Rajya Sabha 2024, 2026) | Gordon and Loeb (2002) | Logical Indian (May 2026) | NTA on NEET CBT shifts (2026)

Every number traces to 71 sources and 16 disclosed assumptions, and no single input tested turns the ₹10,211 Cr base NPV negative

KEY ASSUMPTIONS

ID	Assumption	Value
A01	RRB cycles counted a year	Half a cycle
A02	State exam sittings a year	2 crore
A04	Delay per cancelled recruitment	8 months
A05	Monthly cost of an unfilled post	₹30,000
A06	Household cost per re-sit	₹1,500
A07	Candidate time per re-sit	₹5,000
A08	Re-conduct cost per candidate	₹800
A09	Real discount rate	8%
A10	Coverage, 2027 to 2031	10% to 100%
A14	Probability and blast cuts, base	50% and 60%
A16	Detection odds, now and PRAMAAN	10% and 60%

SENSITIVITY OF BASE NPV (₹ CR): EVERY BAR STAYS ABOVE ZERO

MODEL SUMMARY, BASE CASE (₹ CR)

Line	2027	2031	2036
Expected loss (EAL)	3,317	3,733	4,327
Avoided loss	265	2,986	3,462
PRAMAAN cost	581	674	674
Cumulative net benefit	−316	4,676	17,454

INCIDENTS: STAGE x TYPE

	INS	PHY	CYB	OPS
Creation	4	0	0	0
Storage	10	1	1	0
Access	6	0	0	0
Transport	3	0	0	0
Centre	2	2	0	0
ID check	0	3	0	0
Exam	0	1	7	2
Results	6	0	0	1

SOURCES

Full URLs: research doc, Appendix A (71)

1. Indian Express (2024): 41 leaks, 1.4 crore
2. Wikipedia: leak list; NEET 2024 and 2026
3. NTA, PIB, News On Air: NEET-UG 2026
4. Supreme Court NEET-UG 2024 (SCC Online)
5. PRS: PEA 2024; Amendment 2026
6. Rajya Sabha on NTA (2024, 2026)
7. NTA CBT tender (Mar 2026)
8. Radhakrishnan Committee (Oct 2024)
9. CBI, STF, SIT, EOU via Careers360, ThePrint
10. CERT-In; NIST 800-207; DPDP Rules 2025
11. Kenya KNEC; Korea CSAT; gaokao; SAT
12. Gordon-Loeb; Becker; Shamir; Logical Indian

Source: Team IdeaForge PRAMAAN model (pramaan_model.py) and research document | Full bibliography with URLs in the research document, Appendix A

Thank you

PRAMAAN · Maudrik v7.0 Policy Case · Team IdeaForge

Team Members	Institution	Role
Harsh Kawatra	Delhi Technological University	Team Leader
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